

Alliance justice and relational performance: the mediating role of boundary spanners' citizenship behaviors

Alliance justice
and relational
performance

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Abstract

Purpose – The purpose of this paper is to examine the types of justice that affect knowledge acquisition and opportunism in strategic alliances and how these justice mechanisms function.

Design/methodology/approach – Data were collected from both top-level and operating-level boundary spanners in 295 strategic alliances in China (a total of 590 boundary spanners). A structural equation model (SEM) with bias-corrected bootstrap method was used to test the hypotheses.

Findings – The results show that although both procedural justice and distributive justice are important in deterring opportunism, procedural justice is more effective at enhancing knowledge acquisition than distributive justice is. The results also demonstrate that boundary spanners' helping behaviors are more effective at fostering knowledge acquisition, whereas boundary spanners' voice behaviors have more impact on mitigating opportunism. In addition, boundary spanners' citizenship behaviors partially mediate the relationship between justice and interfirm-level performance.

Originality/value – This study adds a boundary-spanning lens to justice literature by uncovering the missing link between justice and alliance outcomes.

Keywords Boundary spanners, Citizenship behaviors, Distributive justice, Knowledge acquisition, Opportunism, Procedural justice

Paper type Research paper

Introduction

Justice (or fairness), as rooted in equity theory, is fundamental to all types of social and economic exchanges (Liu *et al.*, 2012). The importance of justice has been well established in studies of organizations (e.g. Arab and Atan, 2018; Singh and Singh, 2019; Zhang, 2015). Recent studies have also shown that the importance of justice transcends organizational boundaries and extends to interfirm outcomes (e.g. Luo, 2005; Wang *et al.*, 2014). Although existing literature acknowledges the influence of justice on interfirm transactions, less is understood about the underlying mechanisms through which justice affects exchange outcomes. This research addresses questions concerning the missing link between justice and exchange outcomes in three ways.

First, this paper reasons that the shortfall may be due to the overlooked role of boundary spanners. In strategic alliances, exchange outcomes largely depend on the behaviors of operating-level boundary spanners (Huang *et al.*, 2016; Zaheer *et al.*, 2002) who are directly involved in the exchange process. Meanwhile, justice is perceived and experienced by the boundary spanners who represent alliance partners (Luo, 2005), and boundary spanners' citizenship behavior is one type of reciprocal behavior in response to alliance justice (Van Dyne and LePine, 1998). Therefore, boundary spanners' citizenship behavior may explain the link between alliance justice and performance.

Specifically, we differentiate between two types of citizenship behaviors: helping and voice. Helping behavior refers to boundary spanner behavior that helps an exchange party's



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boundary spanner solve alliance-related problems (Deckop *et al.*, 2003). However, less well understood is another form of citizenship behavior—namely, employee voice (Podsakoff *et al.*, 2009), which refers to constructive suggestions offered by employees with the intent of changing or improving an existing practice or situation (LePine and Van Dyne, 1998, 2001). This paper aims to explore the specific intermediate roles of these two types of boundary spanner behaviors on the relationship between alliance justice and performance.

Second, this paper examines the differing effects of boundary spanners' helping and voice behaviors on two important relational outcomes: knowledge acquisition and opportunism. Alliance boundary spanners are primarily responsible for information exchange and external representation (Aldrich and Herker, 1977). Therefore, they not only facilitate knowledge exchange in alliances (Kawai and Chung, 2019; Nuruzzaman *et al.*, 2019) but also serve to foster cooperation and mitigate exchange hazards in alliance relationships (Huang *et al.*, 2016). Accordingly, we view knowledge acquisition and opportunism reduction as two important outcomes of boundary-spanning behaviors.

Meanwhile, operating-level boundary spanners' helping behavior differs from employee voice in terms of target and controversiality. For example, helping behavior aims to aid fellow boundary spanners from the partner firm; it is intended to benefit other individuals (Podsakoff *et al.*, 2009). Helping behavior is affiliative in nature; it is interpersonal and noncontroversial and provides benefits to the alliance exchange. By contrast, employee voice behavior is directed toward the alliance and enhances the alliance exchange by recommending modifications to exchange procedures or innovative suggestions for change (Van Dyne *et al.*, 2008). Employee voice can be controversial because suggestions for change involve breaking from convention, which may create conflict and negatively affect interpersonal relationships (LePine and Van Dyne, 2001; McAllister *et al.*, 2007). Therefore, boundary spanners' helping and voice behaviors may influence the two alliance outcomes differently.

Finally, although prior research has demonstrated the impact of justice in enhancing interfirm knowledge acquisition and reducing opportunism (Liu *et al.*, 2012; Luo *et al.*, 2015), the effectiveness of different types of justice on the two alliance outcomes may differ. This study extends justice literature by comparing the relative effects of procedural justice and distributive justice on facilitating interfirm knowledge acquisition and mitigating opportunism.

In sum, this study draws upon the justice and boundary-spanning literature and uncovers a missing link between justice and relational performance. Figure 1 shows the conceptual framework. We aim to make three contributions. First, this study adds a boundary-spanning lens to justice literature by examining the intermediate role of individual boundary spanners' behavior in the justice–alliance performance link. Second, this study examines the differing effects of procedural justice and distributive justice on two alliance outcomes. Third, this paper highlights the differences between boundary spanners' helping and voice behaviors and empirically tests the roles of the two types of behaviors in mediating the relationship between justice and the two alliance outcomes.

Theoretical background

Alliance justice: a boundary spanner's perspective

Employees who function across alliance boundaries play a pivotal role in determining the alliance's success (Mäkelä *et al.*, 2019) because alliance strategies and tasks are implemented by operating-level boundary spanners through whom alliance exchanges are operationalized (Janowicz-Panjaitan and Noorderhaven, 2009). However, the effectiveness of those boundary spanners differs in that only a small number of boundary spanners can be viewed by relevant others to facilitate interfirm transactions (Mäkelä *et al.*, 2019). The effectiveness of boundary spanners depends on a variety of factors, including organizational capabilities (e.g. processes, routines) and individual characteristics (e.g. identity, ability, opportunity) (Schotter *et al.*, 2017).

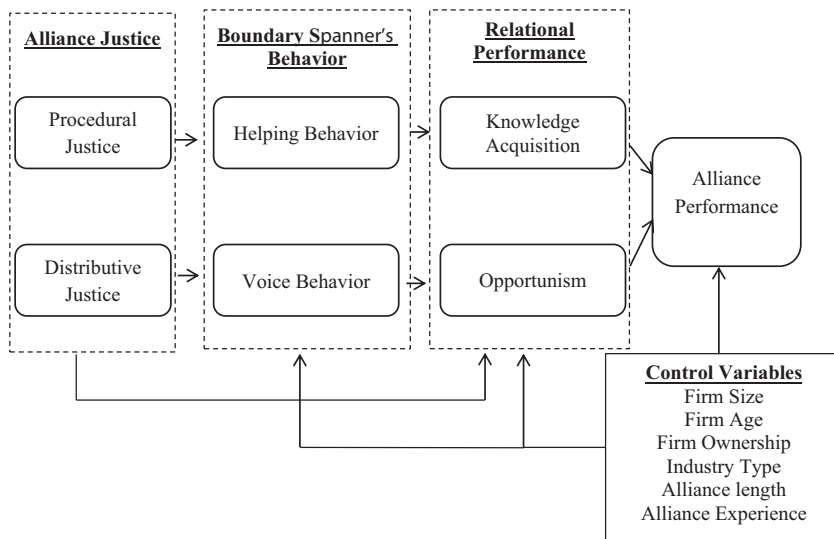


Figure 1.
Conceptual framework

This study focuses on organizational-level factors and examines how alliance justice influences operating-level boundary spanners' behaviors.

Justice is integral to alliance exchanges because it reflects and complements underlying economic forces in guiding interfirm transactions (Luo *et al.*, 2015). According to social exchange theory, reciprocity is probably the most influential exchange rule among norms of social exchange. Alliance justice, as an outcome reflective of organizational processes, promotes feelings of reciprocity among boundary spanners. Reciprocity suggests that if alliance boundary spanners are treated well in the alliance cooperation process, they are highly likely to respond to perceived justice by displaying citizenship behaviors in the alliance, thereby improving task performance and/or reducing counterproductive work behaviors (Singh Sanjay, 2019). Therefore, the fairness of exchange procedures and outcome distribution may influence the attitudes and behaviors of boundary spanners (Ellis *et al.*, 2009; Zhang, 2015).

Hypotheses development

Alliance justice and knowledge acquisition

Procedural justice refers to fairness as perceived by alliance boundary spanners regarding procedures that affect alliance members' interests and gains (Luo, 2005). It emphasizes fairness in decision processes (Zhang, 2015) and is particularly important in influencing interfirm exchanges (Lim and Loosemore, 2017; Luo, 2008). First, procedural justice enhances an alliance boundary spanner's commitment to the alliance tasks (Kim and Mauborgne, 1998), which increases that individual's willingness to share knowledge (Lin, 2007) and influences the final outcome of the alliance relationship (Chaturvedi and Gaur, 2009). Second, knowledge acquisition is more likely to occur in the alliance communication process (Tyler and Bies, 1990). Fair procedures encourage bilateral communication (Chan and Lai, 2017; Luo, 2008) and ensure that valuable knowledge is diffused and shared among alliance boundary spanners (Kim and Mauborgne, 1998).

Distributive justice refers to the perception of fairness in earnings and other outcomes that alliance parties derive from the relationship, such as profit distribution (Poppo and Zhou, 2014). It is founded on the principle of equity. When exchange parties perceive

distributive justice, they recognize the benefit in maintaining the relationship and have high mutual commitment (Kumar *et al.*, 1995). Therefore, they are motivated to transfer knowledge to increase joint performance and advance cooperation. Therefore, we propose the following:

H1. Procedural justice is positively related to knowledge acquisition.

H2. Distributive justice is positively related to knowledge acquisition.

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Alliance justice and opportunism

Opportunism occurs frequently in strategic alliances (Das, 2005). However, when alliance exchanges are buttressed by fair procedures, the opportunity for alliance partners to engage in opportunistic behavior is minimized (Barden *et al.*, 2005). First, fair and routinized procedures increase the likelihood that a partner's opportunistic behavior will be detected (Saxton, 1997) and deterred. In addition, procedural justice fosters bilateral communication and attachment among alliance boundary spanners (Luo, 2008), which further positively influences alliance outcomes (Sambasivan *et al.*, 2011). For example, bilateral communication reduces behavioral uncertainty (Chan and Lai, 2017) and thus restrains opportunism.

We propose that distributive justice also mitigates opportunism in alliances. First, distributive justice enhances the partners' confidence in impartial gain sharing and minimizes the risk of withholding information and shirking obligations (Luo, 2007). Second, boundary spanners tend to treat perception of fairness as a proxy for trust when determining their attitudes and cooperative behaviors in an alliance (Barden *et al.*, 2005). Distributive justice can foster trust that further curtails opportunism. By contrast, when distributive justice is lacking, boundary spanners perceive high work-related uncertainty and ambiguity (Gao, 2017), which may foster opportunistic behavior (Skarlicki *et al.*, 2008). Therefore, we propose the following:

H3. Procedural justice is negatively related to opportunism.

H4. Distributive justice is negatively related to opportunism.

As loosely coupled systems, strategic alliances are confronted with high information and outcome uncertainty. Alliance partners tend to take into account the fairness of the decision-making process when determining their commitment to alliance activities (Luo, 2008). Compared to distributive justice, procedural justice is more likely to foster long-term cooperation because the alliance payoff is difficult to anticipate (Luo, 2008). Based on this line of thinking, this study predicts the following:

H5. Procedural justice in an alliance has a stronger positive effect on (a) enhancing knowledge acquisition and (b) mitigating opportunism than distributive justice does.

Citizenship behaviors and alliance outcomes

Boundary spanners function across unit boundaries and are key to facilitating the interactions between firms and their external environments (Mäkelä *et al.*, 2019; Schotter *et al.*, 2017). Two major roles that boundary spanners play in alliances involve external representation and knowledge exchange (Aldrich and Herker, 1977; Huang *et al.*, 2016). Boundary spanners act as representatives of their organizations by engaging in transaction activities and mitigating exchange hazards in alliances. They are responsible for solving problems during the process of collaboration (e.g. mitigating opportunism or conflict) (Huang *et al.*, 2016). In addition, alliance boundary spanners connect their organization with the counter partner. They process incoming information from alliance partners and share internal information or knowledge externally (Perrone *et al.*, 2003). Therefore, knowledge

acquisition and opportunism reduction are two important outcomes of boundary-spanning behavior. Alliance justice and relational performance

The effectiveness of citizenship behaviors on knowledge acquisition

Although both types of citizenship behaviors can contribute to knowledge acquisition, their effectiveness may differ. Helping behaviors promote interfirm knowledge acquisition in two ways. First, according to the reciprocity norm, an employee's helping behavior *vis-à-vis* coworkers from the partner firm may lead to reciprocal behavior (Gnyawali and Park, 2009). For example, if an employee receives help from an alliance coworker in solving a problem, he or she may then supply knowledge or information when the provider needs it based on the norm of reciprocity (Deckop *et al.*, 2003). Second, helping behaviors foster interpersonal relationships (McAllister *et al.*, 2007), create mutual trust and engender a sense of obligation among individuals (Tang *et al.*, 2008). Mutual trust may increase a knowledge seeker's motivation to learn (Levin and Cross, 2004; Mukherjee *et al.*, 2019) while also strengthening the knowledge provider's motivation to teach. Interpersonal relationships fostered by employees' helping behaviors can also enhance the quality and efficiency of interfirm knowledge transfer (Mukherjee *et al.*, 2019; Zhou *et al.*, 2014).

Conversely, although employee voice behaviors may offer constructive ideas and suggestions for change (Dedahanov *et al.*, 2016), the change orientation is likely to challenge the status quo and affect interpersonal relationships (LePine and Van Dyne, 2001; McAllister *et al.*, 2007). As a result, boundary spanners representing another party in the alliance may not be willing to share their knowledge or information because of disharmony created by voice behaviors. Therefore, this study proposes the following:

- H6. Boundary spanners' helping behaviors have a stronger positive effect on knowledge acquisition in an alliance than boundary spanners' voice behaviors do.

The effectiveness of citizenship behaviors on opportunism

From a transaction cost perspective, the major sources of opportunism are exchange characteristics such as asset specificity and uncertainty (Williamson, 1985). Contracts serve as a mechanism to safeguard against opportunism (Bai *et al.*, 2016; Zhang *et al.*, 2018). Employee voice behaviors are targeted at alliance transactions, and boundary spanners' voice behaviors can enable firms to refine interfirm contracts when suggestions for improving cooperative procedures lead to modification of predefined contract agreements (Van Dyne and LePine, 1998; Zhang *et al.*, 2018). When provisions in the original contract do not accommodate new developments or adapt to dynamics in the environment, modifying inappropriate exchange procedures is necessary to mitigate a partner's opportunistic behaviors (Ghosh and John, 2009; Zhang *et al.*, 2018). Meanwhile, employee voice behaviors, such as pointing out a partner's proscribed behaviors and suggesting appropriate alternatives, may deter the partner's unethical behaviors.

By contrast, boundary spanners' helping behaviors are directed toward individuals (Podsakoff *et al.*, 2009) and help create harmonious interpersonal relationships among operating-level boundary spanners. However, the goodwill generated by boundary spanners may only accrue for those with whom they have direct contact and may not escalate to become organizational bonding. That means that boundary spanners' helping behaviors may not directly influence interorganizational exchanges (Zaheer *et al.*, 1998). Theoretically, opportunism as an interfirm outcome is primarily suppressed by interfirm governance (e.g. Williamson, 1985; Zhou and Poppo, 2010). Therefore, this paper proposes the following:

- H7. Boundary spanners' voice behaviors have a stronger negative effect on opportunism than boundary spanners' helping behaviors do.

Methods

Sampling and data collection

To test these hypotheses, we collected data through a field study and a large-scale survey. We randomly selected a sample of 500 firms with strategic alliances from a directory provided by a marketing research firm located in Shenzhen, China. Top executives of the firms served as the first key informants. Well-trained interviewers telephoned these top executives and obtained oral agreements from 352 of them to participate. Of the 352 firms contacted, we successfully interviewed 311 firms. Following the interview with the top-level alliance boundary spanners, we asked each of them to set up a meeting with an operating-level alliance boundary spanner who was responsible for a specific project with the alliance partner. Sixteen samples had extensive missing values, so the final sample consisted of 295 firms with two matched informants, a response rate of 59.0% (295/500). The samples span various industries. The average duration of the strategic alliances was 6.01 years. The focal firms' size in terms of employee head count ranged from fewer than 30 to more than 40,000, and annual sales varied from less than RMB 1,200,000 to more than RMB 20,000,000.

Measures

We adapted measures of procedural justice and distributive justice from [Liu et al. \(2012\)](#) by refining the four items to fit into our alliance context. The measure of employee helping behavior was adapted from [Podsakoff et al. \(1990\)](#), with four items capturing alliance boundary spanners' willingness to provide assistance to alliance partners. We adapted the measure of employee voice behavior from [LePine and Van Dyne \(1998\)](#). These three items capture the extent to which alliance boundary spanners provide recommendations and communicate their opinions in alliance exchanges. From [Muthusamy and White \(2005\)](#), we adapted the measure of knowledge acquisition, which assesses the amount of knowledge, skill and competency that alliance partners absorb from their alliance exchanges. The measure of items of opportunism assesses the extent to which alliance partners engage in behaviors that are "self-interest seeking with guile" ([Williamson, 1985](#)). We adopted the six items from [Heide et al. \(2007\)](#). The five items to measure alliance performance were adapted from [Lambe et al. \(2002\)](#) and [Becerra et al. \(2008\)](#).

We also controlled extraneous variables that might affect alliance performance, including previous alliance experience, firm age, firm size, firm ownership (0 = foreign invested, 1 = local firm), industry type (0 = service industries, 1 = manufacturing industries) and alliance length.

Measurement validity

This study adopted confirmatory factor analysis to assess the validity and reliability of the multi-item constructs. Overall, the measurement model exhibited acceptable fit indices ($\chi^2_{(402)} = 584.96, p < 0.001$; CFI = 0.97; IFI = 0.97; NFI = 0.90; RMSEA = 0.04). The composite reliability and average variance extracted (AVE) for all constructs are greater than the 0.70 and 0.50 thresholds, respectively (see the [Appendix](#)). This suggests that the measures have satisfactory convergent validity.

We assessed the discriminant validity of the constructs by comparing the AVE for the individual constructs with the shared variance between all possible pairs of constructs. The results showed that the AVE for each construct was higher than its highest shared variance with other variables, confirming discriminant validity ([Fornell and Larcker, 1981](#)). [Table 1](#) shows the details.

Common method bias

To mitigate common method bias (CMB), we collected data from two key informants in each alliance and empirically assessed the influence of potential CMB using a marker variable

	1	2	3	4	5	6	7	8	9	10	11	12	13
1 Procedural justice													
2 Distributive justice	0.171**												
3 Knowledge acquisition	0.512**	0.165**	0.509**	-0.251**	0.326**	0.244**	0.256**	-0.066	-0.098	0.090	-0.079	-0.073	0.199**
4 Opportunism	-0.242**	0.099	0.093	-0.247**	0.021	0.265**	0.200**	-0.078	-0.168**	0.080	-0.023	-0.143*	0.231**
5 Helping behavior	0.330**	-0.238**	-0.179**	-0.187**	0.363**	0.104	0.282**	-0.022	-0.030	0.100	-0.006	-0.118*	0.184**
6 Voice behavior	0.249**	0.028	0.367**	-0.103	-0.111	-0.289**	-0.218**	-0.005	0.061	-0.049	0.034	0.084	-0.131*
7 Alliance performance	0.261**	0.270**	0.110	-0.280**	0.173**	0.167**	0.217**	-0.073	-0.031	0.061	-0.051	-0.130*	0.125*
8 Firm size	-0.059	-0.070	-0.015	0.002	0.222**	0.090	0.084	-0.141*	-0.177**	0.039	-0.045	-0.239**	0.098
9 Firm age	-0.090	-0.160**	-0.023	0.068	-0.065	-0.133*	0.064	0.412**	0.023	0.043	0.173**	-0.019	0.570**
10 Firm ownership	0.096	0.086	0.107	-0.042	0.068	-0.169**	0.030	0.147*	0.408**	0.141*	0.268**	0.092	0.069
11 Alliance length	-0.071	-0.016	0.001	0.041	-0.044	-0.038	0.179**	0.273**	0.131*	0.125*	0.562**	0.053	0.138*
12 Industry type	-0.065	-0.135*	-0.110	0.090	-0.122*	-0.230**	-0.012	0.098	0.565**	0.158**	0.152**	-0.208**	0.061
13 Alliance experience	0.205**	0.236**	0.190**	-0.123*	0.131*	0.104	0.573**	0.076	0.060	-0.200**	-0.051	-0.058	0.159**
14 MV Marker	-0.148*	-0.113	-0.015	0.113	0.097	-0.134*	-0.074	0.089	0.144*	0.068	0.165**	0.022	0.007
Mean	5.411	5.319	5.263	2.746	5.250	5.019	5.671	4.996	2.321	0.315	6.020	1.492	5.585
Standard Deviation	0.824	1.321	0.874	1.333	0.972	1.378	0.724	1.430	0.693	0.465	4.304	0.501	0.820

Note(s): * $p < 0.05$; ** $p < 0.01$ (two-tailed); procedural justice, knowledge acquisition, opportunism, helping and voice behavior are reported by the operating-level boundary spanners, whereas all other variables are reported by the top-level boundary spanners. $N = 295$

Zero-order correlations are below the diagonal; adjusted correlations for the potential common method variance (Lindell and Whitney, 2001) are above the diagonal

Table 1.
Descriptive statistics
and correlations

approach (Lindell and Whitney, 2001). A five-item scale that measured the operating-level of boundary spanners' perceived risk of online shopping (Cronbach's $\alpha = 0.90$) was selected as the marker variable because it is theoretically unrelated to at least one key construct in this study. After the partial correlation adjustment, all the significant correlations remained significant (see Table 1), suggesting that CMB was unlikely to be a concern in this study (Lindell and Whitney, 2001).

Analysis and results

This study employs a structural equation model (SEM) with the maximum likelihood estimation method to test H1 through H7. The study also follows Cheung and Lau's (2008) method and uses the BC bootstrap method to examine confidence intervals for the mediation effect (Cheung and Lau, 2008).

The results of the full SEM provide a satisfactory model fit ($\chi^2(562) = 903.658$, [CFI] = 0.942, [IFI] = 0.943, [RMSEA] = 0.045). Figure 2 presents the path coefficients for the full model. In line with H1 and H3, procedural justice positively influences knowledge acquisition ($\beta = 0.535$, $p < 0.01$), and it has a significant negative effect on opportunism ($\beta = -0.164$, $p < 0.05$). The effect of distributive justice on knowledge acquisition in alliances is insignificant ($\beta = 0.006$, n.s.); thus, H2 is not supported. In line with H4, distributive justice negatively influences opportunism ($\beta = -0.147$, $p < 0.05$). In H5a/ H5b, this study predicts that procedural justice is more effective than distributive justice at fostering knowledge acquisition and in mitigating opportunism. The t -test for the equality of the regression coefficients supports H5a ($t = 6.537$, $p < 0.01$) but does not support H5b ($t = 0.114$, n.s.).

Consistent with H6, boundary spanners' helping behaviors positively affect knowledge acquisition in alliances ($\beta = 0.203$, $p < 0.01$), and the effect of voice behaviors on knowledge acquisition is insignificant ($\beta = -0.098$, n.s.). The t -test ($t = 4.395$, $p < 0.01$) for the equality of the two regression coefficients also shows that they are statistically different. Consistent with H7, boundary spanners' voice behaviors negatively influence opportunism ($\beta = -0.216$, $p < 0.01$), and the effect of helping behavior on opportunism is insignificant ($\beta = -0.012$, n.s.). The t -test for the equality of the two regression coefficients ($t = 1.835$, $p < 0.1$) indicates a marginally significant difference between them; thus, H7 is supported. This study summarizes the results of the hypotheses testing in Table 2.

Post hoc analyses on the mediating role of boundary spanners' citizenship behaviors

The researchers further explored the mediating impact of boundary spanners' citizenship behaviors on the relationship between justice and alliance outcomes. As Cheung and Lau (2008) recommended, the researchers tested the mediating effect by employing the bootstrap estimation of 95th percentile confidence intervals with 1,000 samples. From Table 3, it can be seen that the 95% confidence interval bootstrap estimates for the indirect effect of boundary spanners' helping behaviors on procedural justice and knowledge acquisition link, the indirect effect of boundary spanners' voice behaviors on procedural justice and opportunism link and the indirect effect of boundary spanners' voice behaviors on distributive justice and opportunism link do not include zero, suggesting a significant mediation effect.

Discussions

Theoretical contributions

Extant studies have examined the relationship between justice and alliance performance at the firm level. First, this study adds a boundary-spanning lens to alliance exchanges and offers new insights into how alliance justice influences relational performance through

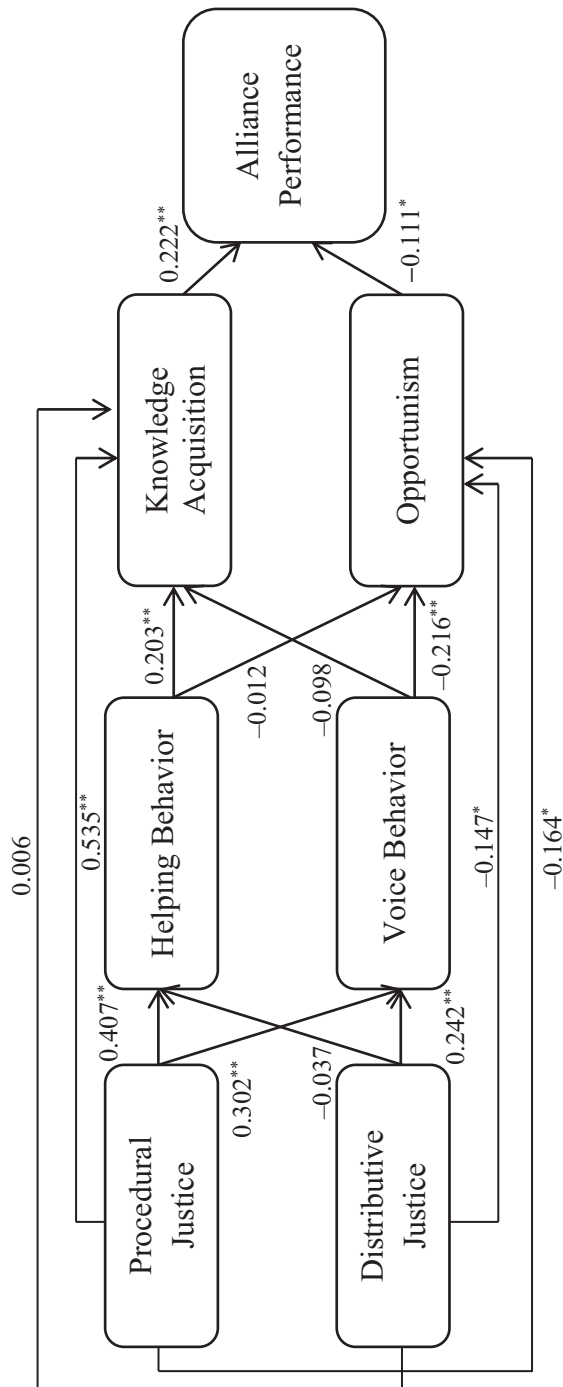


Figure 2.
Structural equation
model with path
coefficients

Table 2.
Results of hypotheses
testing

No	Research hypotheses	β or t -value	p -value	Research hypotheses supported/not supported
H1	Procedural justice is positively related to knowledge acquisition	$\beta = 0.535$	$p < 0.01$	Supported
H2	Distributive justice is positively related to knowledge acquisition	$\beta = 0.006$	n.s.	Not supported
H3	Procedural justice is negatively related to opportunism	$\beta = -0.164$	$p < 0.05$	Supported
H4	Distributive justice is negatively related to opportunism	$\beta = -0.147$	$p < 0.05$	Supported
H5a	Procedural justice in an alliance has a stronger positive effect on enhancing knowledge acquisition than distributive justice does	$t = 6.537$	$p < 0.01$	Supported
H5b	Procedural justice in an alliance has a stronger positive effect on mitigating opportunism than distributive justice does	$t = 0.114$	n.s.	Not supported
H6	Boundary spanners' helping behaviors have a stronger positive effect on knowledge acquisition in an alliance than boundary spanners' voice behaviors do	$t = 4.395$	$p < 0.01$	Supported
H7	Boundary spanners' voice behaviors have stronger negative effect on opportunism than boundary spanners' helping behaviors do	$t = 1.835$	$p < 0.10$	Weakly supported

Table 3.
Mediation analysis via
bootstrapping

	Direct without mediator	Direct with mediator	Indirect effect	Bootstrapping bias-corrected 95% CI		Significance
Procedural Justice → Helping → Knowledge Acquisition	0.588**	0.517**	0.082	0.020	0.181	0.010
Procedural Justice → Helping → Opportunism	-0.227**	-0.190*	-0.003	-0.065	0.056	0.864
Procedural Justice → Voice → Knowledge Acquisition	0.588**	0.553**	-0.030	-0.082	0.004	0.082
Procedural Justice → Voice → Opportunism	-0.227**	-0.162*	-0.063	-0.131	-0.019	0.006
Distributive Justice → Helping → Knowledge Acquisition	n.s.	n.s.	-0.009	-0.048	0.016	0.430
Distributive Justice → Helping → Opportunism	-0.189**	-0.154**	0.001	-0.007	0.017	0.649
Distributive Justice → Voice → Knowledge Acquisition	n.s.	n.s.	-0.024	-0.070	0.004	0.089
Distributive Justice → Voice → Opportunism	-0.189**	-0.147*	-0.052	-0.126	-0.013	0.006

Note(s): Standardized estimates are shown. 1,000 bootstrapping samples were used. Two tailed significance. CI = Confidence interval. * $p < 0.05$; ** $p < 0.01$

boundary spanners' citizenship behaviors. Our results show that boundary spanners' citizenship behaviors partially mediate the relationship between justice and relational performance. As suggested by previous studies (Mäkelä *et al.*, 2019; Nuruzzaman *et al.*, 2019), the findings of this study further confirm the importance of boundary spanners in alliances.

Second, this study adds support to the conventional view that justice is fundamental to all types of economic and social transactions. More important, it answers the question about which justice type is more effective at enhancing knowledge acquisition and mitigating opportunism in strategic alliances. The results demonstrate that procedural justice is more effective at enhancing knowledge acquisition in alliances than distributive justice is.

Third, this study adds new insight into the citizenship behavior literature by exploring the different roles that boundary spanners' helping and voice behaviors play in influencing relational performance in alliances. The results show that boundary spanners' helping behaviors play a more important role in fostering knowledge acquisition than voice behaviors do because helping behaviors are interpersonal (McAllister *et al.*, 2007) and motivate boundary spanners to share and receive knowledge. By contrast, voice behaviors may upset interpersonal relationships because they challenge the status quo and reduce boundary spanners' willingness to share knowledge. However, boundary spanners' voice behaviors are more influential at mitigating opportunism than helping behaviors are. This study reasons that voice behaviors may help exchange parties improve formal contracts (Van Dyne and LePine, 1998), which serve as a primary mechanism to safeguard against opportunism. Yet the contributions of interpersonal relationships fostered by helping behaviors are directed toward individuals (Podsakoff *et al.*, 2009) and may not escalate to organizational bonding; thus, they may have limited influence in mitigating opportunism.

Managerial implications

The findings of this study offer several important managerial implications for organizations in emerging markets. First, this study confirms the importance of alliance justice in strategic alliance exchanges and further suggests that managers should recognize the important roles of procedural and distributive justice in improving alliance performance.

Second, this research shows that the impact of alliance justice on relational performance operates through boundary spanners' citizenship behaviors. This indirect effect suggests that to facilitate alliance transactions, exchange parties may choose to develop fair exchange procedures and equitable outcome distribution policies or to encourage boundary spanners' citizenship behaviors or both. Organization and alliance managers should recognize the importance of operating-level boundary spanners because they are primarily responsible for daily alliance exchanges and operating efficiency (Ireland *et al.*, 2002).

Third, managers should bear in mind that boundary spanners' citizenship behaviors have significant influence on interfirm knowledge acquisition and opportunism. Therefore, managers are advised to promote various types of citizenship behaviors to achieve cooperation outcomes. For example, if alliance managers want to boost knowledge acquisition in an alliance, they should promote operating-level boundary spanner helping behaviors.

Limitations and further research

This study has several limitations that may prompt further studies. First, the cross-sectional design of this research prevented the study from testing a causal relationship in the model. Additional longitudinally designed studies are suggested to ascertain causality patterns. Second, future studies might explore the impact of justice on different types of boundary spanner citizenship behaviors. This study focused only on boundary spanners' helping and voice behaviors to keep the model parsimonious. Further studies might examine whether

procedural and distributive justice affects types of citizenship behaviors differently and whether different types of boundary spanners' citizenship behaviors influence knowledge acquisition and opportunism in alliances differently.

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Appendix
Measures

Construct and source	Description	Factor loadings
Procedural Justice (Liu <i>et al.</i> , 2012) CR = 0.832; AVE = 0.554	(1) This alliance partner does not discriminate but treats us impartially	0.707
	(2) This alliance partner has used consistent policies and decision-making procedures in the alliance exchanges	0.797
	(3) This alliance partner always takes into account our concern and feedback in its policies and programs	0.733
	(4) This alliance partner knows well the situations we face and explains to us its alliance-related decisions	0.738
Distributive Justice (Liu <i>et al.</i> , 2012) CR = 0.943; AVE = 0.806	(1) Our gain from this alliance is consistent with the amount of effort and investment we have made	0.919
	(2) Our gain from this alliance is commensurate with the role and responsibilities we have actually taken	0.916
	(3) Our gain relative to our contribution from this relationship is about the same as that for our alliance partner	0.931
	(4) Our gain is proportionate to our performance in all joint activities undertaken by both partners	0.820
Employee Helping Behavior (Podsakoff <i>et al.</i> , 1990) CR = 0.870; AVE = 0.626	(1) Help alliance partners who have been absent	0.725
	(2) Help alliance partners who have heavy work loads	0.820
	(3) Help orient new colleagues from both organization and alliance partner	0.804
	(4) Willingly help alliance partners who have work-related problems	0.813
Employee Voice Behavior (LePine and Van Dyne, 1998) CR = 0.831; AVE = 0.713	(1) I seldom develop and make recommendations concerning issues that affect this strategic alliance ^R	0.751
	(2) I will keep silent and not communicate my opinions about work-related issues to an alliance partner when his or her opinion is different from mine ^R	0.928
	(3) I will try to keep well-informed about issues where my opinion might be useful to this alliance	deleted
	(4) I will try to keep well-informed about issues where my opinion might be useful to this alliance	deleted
Knowledge Acquisition (Muthusamy and White, 2005) CR = 0.823; AVE = 0.610	(1) Our firm has learned to jointly execute marketing, R&D or production operations with this alliance partner	0.710
	(2) Our firm has learned to exchange skills, know-how or technologies with this alliance partner	0.866
	(3) Our firm has gained new techniques, competencies or technologies from this alliance partner	0.760

(continued)

Construct and source	Description	Factor loadings
Opportunism (Heide <i>et al.</i> , 2007) CR = 0.936; AVE = 0.711	(1) On occasion, we lie about certain things to protect our interests	0.696
	(2) We sometimes promise to do things without actually doing them	0.824
	(3) We do not always act in accordance with our contract(s)	0.809
	(4) We sometimes try to breach informal agreements between our companies to maximize our own benefit	0.899
	(5) We will try to take advantage of “holes” in our contract to further our own interests	0.905
	(6) We sometimes use unexpected events to extract concessions from the alliance partner	0.906
Alliance Performance (Becerra <i>et al.</i> , 2008; Lambe <i>et al.</i> , 2002) CR = 0.840; AVE = 0.512	(1) We and this partner have achieved a high level of joint profits	0.733
	(2) We and this partner have generated a lot of profits together	0.680
	(3) We and this partner have increased joint profits	0.648
	(4) Our cooperation with this partner can be characterized as a success	0.745
	(5) Our cooperation with this partner has contributed to growth in our firm	0.766
Previous Alliance Experience (Lunnan and Haugland, 2008) CR = 0.826; AVE = 0.615	(1) Our firm is experienced in interfirm collaboration	0.770
	(2) We have accumulated a lot of experience in alliance management	0.885
	(3) We have learned how to handle alliance collaboration through previous experience of corporate cooperation	0.684

Note(s): All items use Likert scales (1 = strongly disagree; 7 = strongly agree). CR = composite reliability
^R represents items that have been reverse coded

Measurement model fit: χ^2 (402) = 584.959, $p < 0.001$; CFI = 0.968; IFI = 0.968; NFI = 0.904; RMSEA = 0.039

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